Communication Systems Lecture-5: Pulse Amplitude Modulation

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Introduction

Pulse Amplitude Modulation (PAM): corresponds to the conversion of an analog signal to a pulse-type signal in which the amplitude of the pulse denotes the analog information.
There are two classes of PAM signals:

- Natural Sampling (or gating):
  - Easy to generate.
- Instantaneous Sampling that produces flat-top pulses:
  - Useful for analog-to-digital conversion.

Natural Sampling (1)

If $w(t)$ is an analog signal bandlimited to B Hz, the PAM signal that uses natural sampling is:

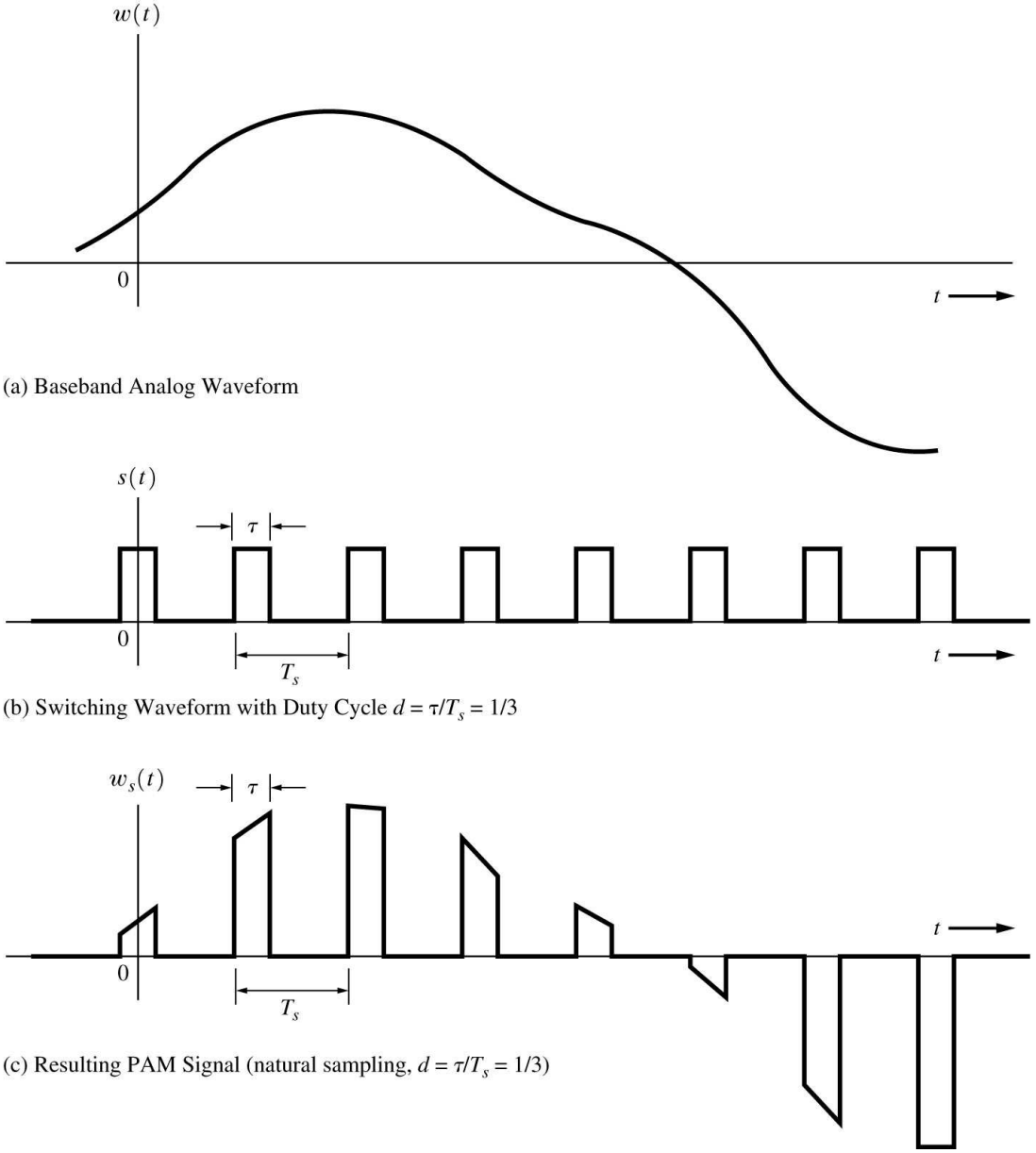
$$\begin{aligned} w_s(t) &= w(t)s(t) \\ &\triangleq w(t) \sum_{k=-\infty}^{+\infty} \mathrm{rect}\left(\frac{t-kT_s}{\tau}\right) \end{aligned}$$

where $f_s = \frac{1}{T_s} \geq 2B$.

- The duty cycle of the rectangular wave switching waveform $s(t)$ is:

$$d = \frac{\tau}{T_s} < 1$$

Natural Sampling (2)



(a) Baseband Analog Waveform

(b) Switching Waveform with Duty Cycle $d = \tau/T_s = 1/3$

(c) Resulting PAM Signal (natural sampling, $d = \tau/T_s = 1/3$)

Natural Sampling (3)

- Since $w_s(t) = w(t)s(t)$, the Fourier transform of $w_s(t)$ can be calculated from:

$$W_s(f) = \mathcal{F}[w_s(t)] = W(f) * S(f)$$

- Since $s(t)$ is periodic (with period T_s), it has the following Fourier series representation:

$$s(t) = \sum_{n \in \mathbb{Z}} c_n e^{j2\pi n f_s t}$$

- The Fourier coefficients can be calculated from:

$$\begin{aligned} c_n &= \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} s(t) e^{-j2\pi n f_s t} \mathrm{d}t \\ &= \frac{1}{T_s} \int_{-\tau/2}^{\tau/2} e^{-j2\pi n f_s t} \mathrm{d}t = \frac{1}{-j2\pi n f_s T_s} \left[e^{-j2\pi n f_s t} \right]_{-\tau/2}^{\tau/2} \\ &= \frac{1}{j2\pi n} \left[e^{j\pi n f_s \tau} - e^{-j\pi n f_s \tau} \right] \end{aligned}$$

Natural Sampling (4)

- Since $f_s\tau = \tau/T_s = d$, then:

$$c_n = \frac{1}{j2\pi n}\left[e^{j\pi nd} - e^{-j\pi nd}\right] = \frac{d}{j2\pi nd}\left[e^{j\pi nd} - e^{-j\pi nd}\right]$$
$$= \frac{d}{\pi nd}\sin(\pi nd) = d\,\mathrm{sinc}\,(nd)$$

- Finally:

$$s(t) = \sum_{n\in\mathbb{Z}} d\,\mathrm{sinc}\,(nd)\, e^{j2\pi nf_s t}$$

- This implies that the Fourier transform of $s(t)$ is given by:

$$S(f) = d\sum_{n\in\mathbb{Z}} \mathrm{sinc}\,(nd)\,\delta(f - nf_s)$$

Natural Sampling (5)

Finding $S(f)$, second method:

- It was possible to evaluate $S(f)$ as follows:

$$
\begin{aligned}
s(t) &= \mathrm{rect}\left(\frac{t}{\tau}\right) * \frac{1}{T_s}\mathrm{comb}\left(\frac{t}{T_s}\right) \\
\Rightarrow \quad S(f) &= \tau\mathrm{sinc}(\tau f)\mathrm{comb}(T_s f) \\
&= \tau\mathrm{sinc}(\tau f)\frac{1}{T_s}\sum_{n\in\mathbb{Z}}\delta(f - nf_s) \\
&= \frac{\tau}{T_s}\sum_{n\in\mathbb{Z}}\mathrm{sinc}(\tau n f_s)\delta(f - nf_s) \\
&= d\sum_{n\in\mathbb{Z}}\mathrm{sinc}\left(nd\right)\delta(f - nf_s)
\end{aligned}
$$

since $\tau/T_s = \tau f_s = d$.

Natural Sampling (6)

- Consequently:

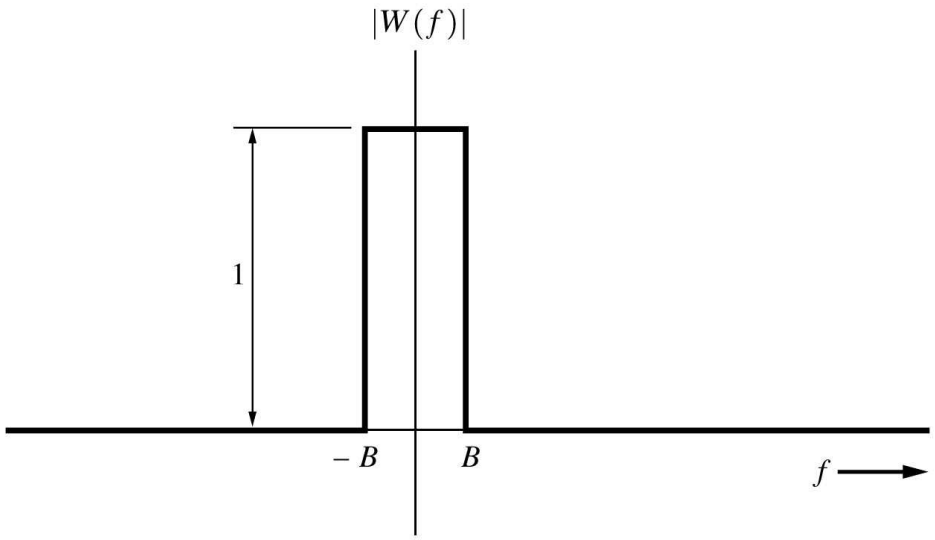
$$
\begin{aligned}
W_s(f) &= W(f) * d \sum_{n\in\mathbb{Z}} \operatorname{sinc}(nd)\,\delta(f - nf_s) \\
&= d \sum_{n\in\mathbb{Z}} \operatorname{sinc}(nd)\,[W(f) * \delta(f - nf_s)] \\
&= d \sum_{n\in\mathbb{Z}} \operatorname{sinc}(nd)\,W(f - nf_s)
\end{aligned}
$$

- As with impulse sampling, the replicas of $W(f)$ do not interfere with each other when $f_s \geq 2B$ (since two consecutive replicas are separated by f_s.)
- In this case:

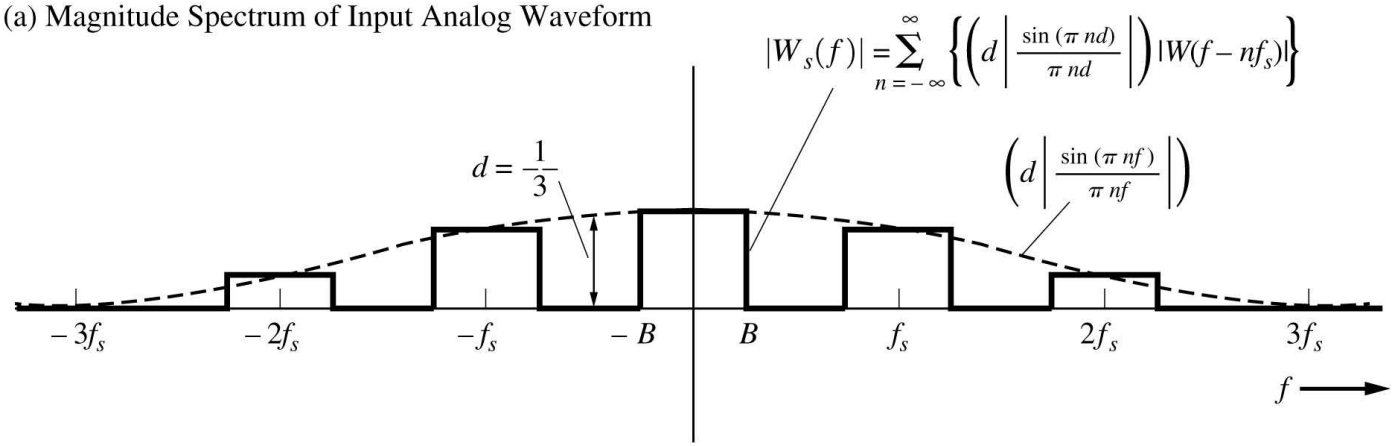
$$
|W_s(f)| = d \sum_{n\in\mathbb{Z}} |\operatorname{sinc}(nd)|\,|W(f - nf_s)|
$$

Natural Sampling (7)

- The next figure shows $|W_s(f)|$ for $f_s = 4B$ and $d = 1/3$.



(a) Magnitude Spectrum of Input Analog Waveform



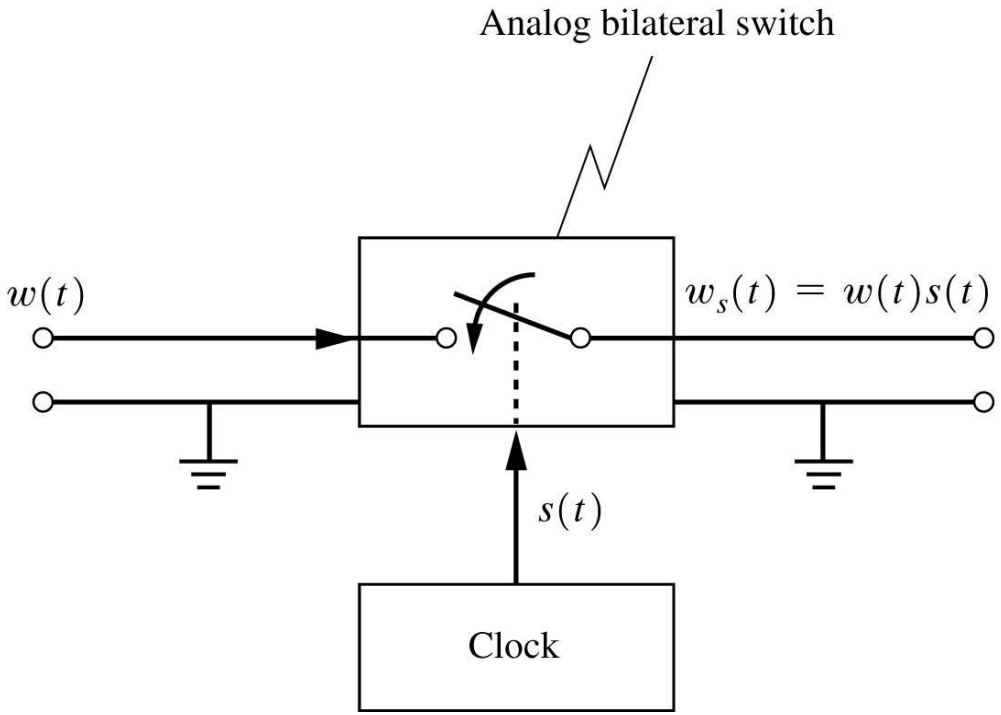
(b) Magnitude Spectrum of PAM (natural sampling) with $d = 1/3$ and $f_s = 4$ B

Natural Sampling (8)

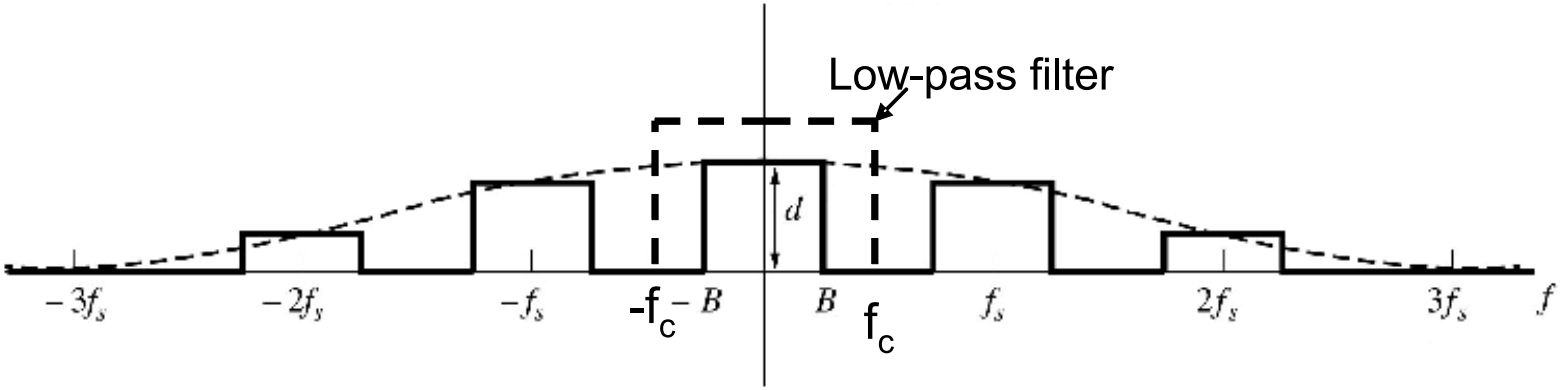
- Note that $\mathrm{sinc}\,(nd) = 0$ when n is an integer multiple of $1/d$.This implies that there are no replicas of $W(f)$ around frequencies that are multiples of f_s/d.
- From the last figure, the first zero is at $3f_s$ implying that the null-to-null bandwidth of the PAM signal (with natural sampling) is $3f_s = 12B$ which is 12 times the bandwidth of the original analog signal (in this example).
- In the general case, the null-to-null bandwidth of a PAM signal is f_s/d (note that $d < 1$).

Natural Sampling (9)

- The PAM waveform with natural sampling is relatively easy to generate since it only requires the use of an analog switch.



Natural Sampling (10)



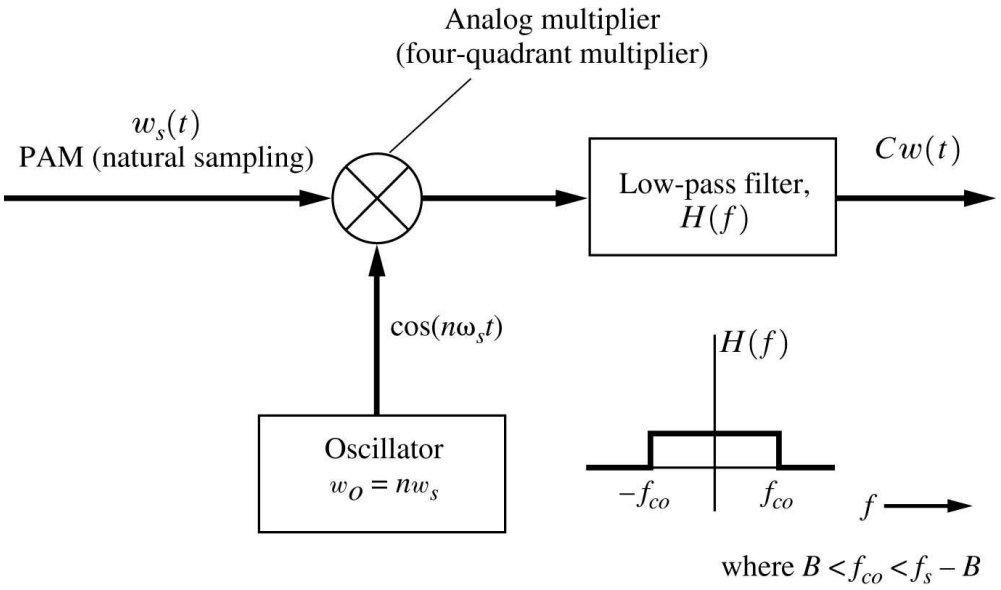
- From the spectrum of the PAM signal, it can be directly seen that the original analog signal $w(t)$ can be recovered from the PAM signal $w_s(t)$ by passing the PAM signal through a low-pass filter whose cutoff frequency verifies:

$$B < f_c < f_s - B$$

- Note that the recovered spectrum would be identical to the original spectrum except for a gain factor that is equal to d.
  - This can be simply compensated for by the use of an amplifier ($d < 1$).

Natural Sampling (11)

- Note that if significant noise is present around $f = 0$, the following detector can be used.

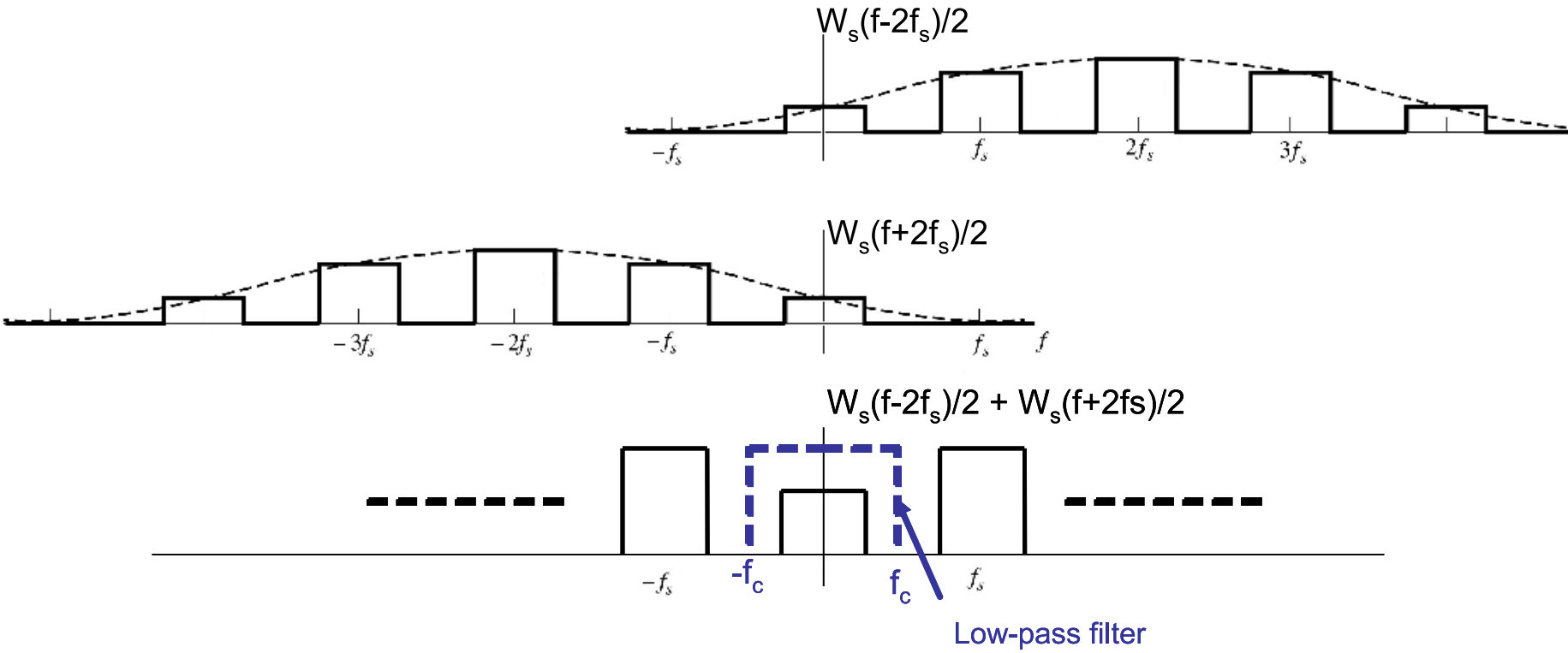


- This shifts the PAM signal that was centered about nf_s to baseband ($f = 0$).
- The spectrum of the signal at the output of the multiplier is:

$$\frac{1}{2}W_s(f - nf_s) + \frac{1}{2}W_s(f + nf_s)$$

Natural Sampling (12)

- Example: for $n = 2$:



Instantaneous Sampling (1)

If $w(t)$ is an analog signal bandlimited to B Hz, the instantaneous sampled PAM signal is given by:

$$w_s(t) = \sum_{k=-\infty}^{+\infty} w(kT_s)h(t - kT_s)$$

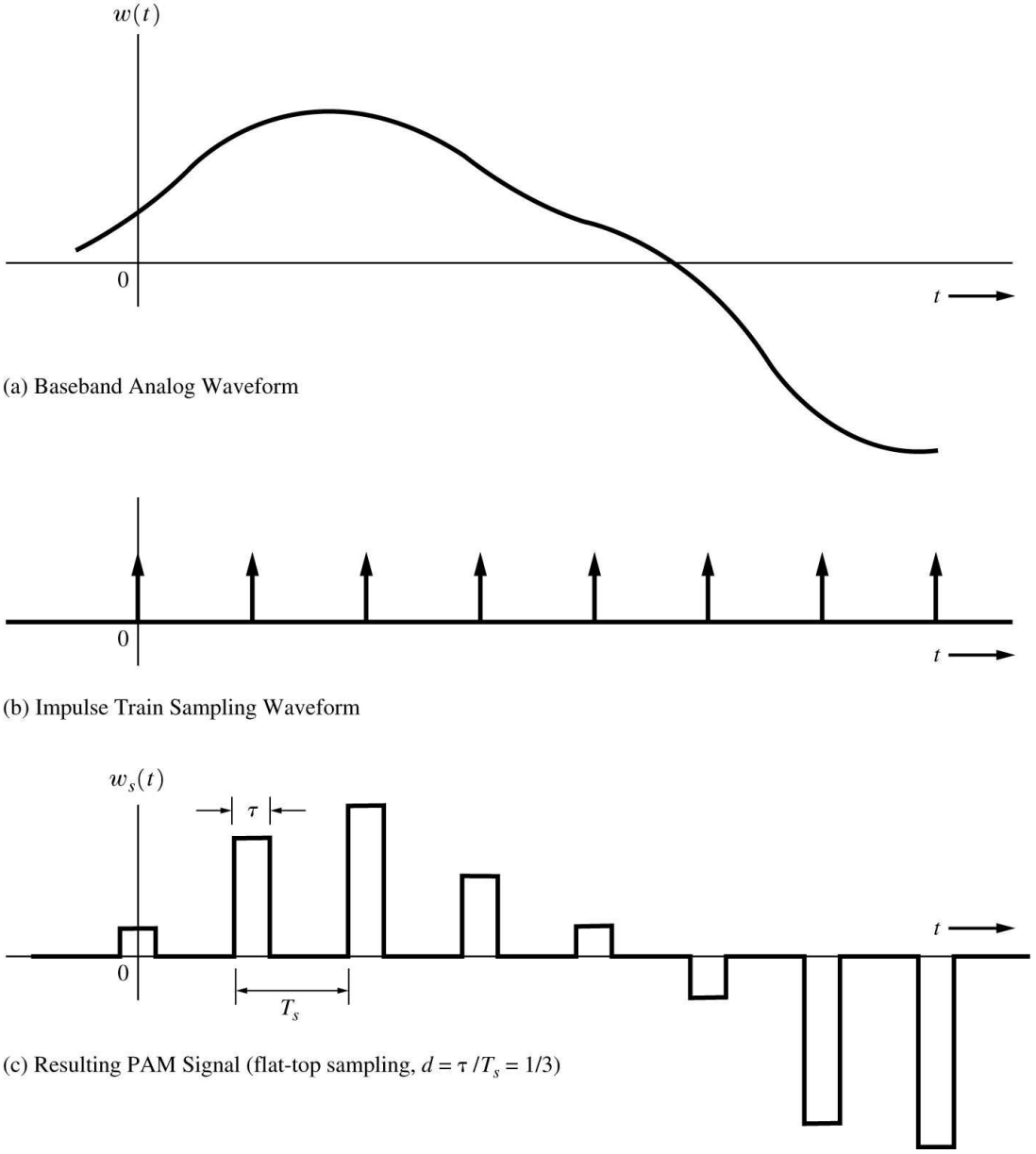
where $h(t)$ denotes the sampling-pulse shape. For flat-top sampling:

$$h(t) = \mathrm{rect}\left(\frac{t}{\tau}\right)$$

where $f_s \geq 2B$ and the duty cycle is $d = \tau/T_s < 1$.

- Note that another pulse shapes can be used. When $h(t)$ is a sinc-type function, the modulation equation becomes identical to the sampling theorem.

Instantaneous Sampling (2)



Instantaneous Sampling (3)

- The flat-top PAM modulated signal can be written as:

$$
\begin{aligned}
w_s(t) &= \sum_k w(kT_s)\left[h(t) * \delta(t - kT_s)\right] \\
&= h(t) * \sum_k w(kT_s)\delta(t - kT_s) \\
&= h(t) * \left[w(t)\sum_k \delta(t - kT_s)\right]
\end{aligned}
$$

- Consequently:

$$
\begin{aligned}
W_s(f) &= H(f)\left[W(f) * \mathcal{F}\left[\sum_k \delta(t - kT_s)\right]\right] \\
&= H(f)\left[W(f) * \frac{1}{T_s}\sum_k \delta(f - kf_s)\right] \\
&= \frac{1}{T_s}H(f)\sum_k W(f - kf_s)
\end{aligned}
$$

Instantaneous Sampling (4)

- As with impulse sampling and naturally-sampled PAM, the overlap between the replicas of $W(f)$ can be avoided when $f_s \geq 2B$.
- In this case:

$$|W_s(f)| = \frac{1}{T_s}|H(f)|\sum_k |W(f - kf_s)|$$

- When $h(t) = \mathrm{rect}\left(\frac{t}{\tau}\right)$, $H(f) = \tau \mathrm{sinc}\left(\tau f\right)$ and:

$$|W_s(f)| = \frac{\tau}{T_s}|\mathrm{sinc}\left(\tau f\right)|\sum_k |W(f - kf_s)|$$

- The zero frequency crossings occur at:

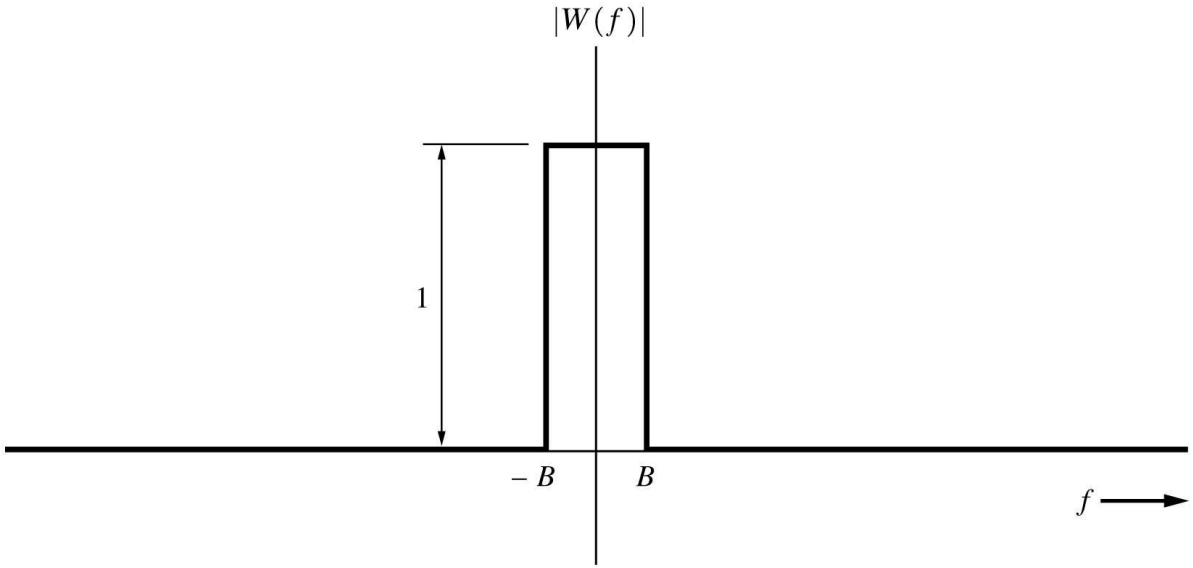
$$f = n\frac{1}{\tau} \quad ; \quad n \in \mathbb{Z}$$

$$\Rightarrow \quad f = n\frac{1}{d}f_s \quad ; \quad n \in \mathbb{Z}$$

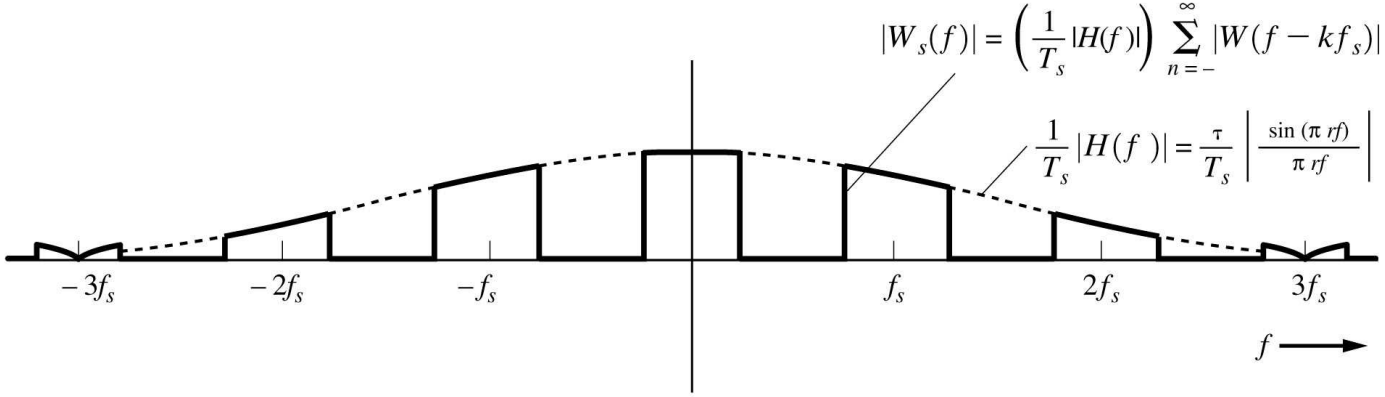
since $d = \frac{\tau}{T_s} = \tau f_s$.

Instantaneous Sampling (5)

- The next figure shows $|W_s(f)|$ for $f_s = 4B$ and $d = 1/3$.



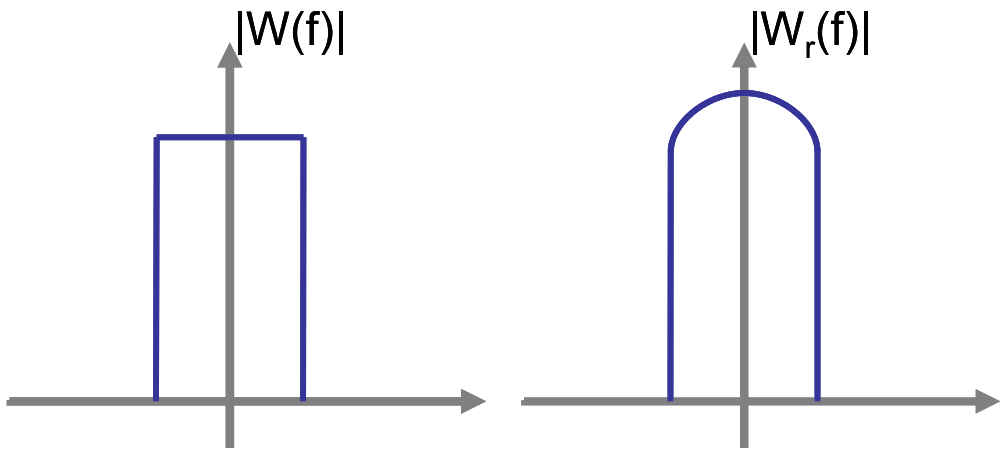
(a) Magnitude Spectrum of Input Analog Waveform



(b) Magnitude Spectrum of PAM (flat-top sampling), $\tau/T_s = 1/3$ and $f_s = 4B$

Instantaneous Sampling (6)

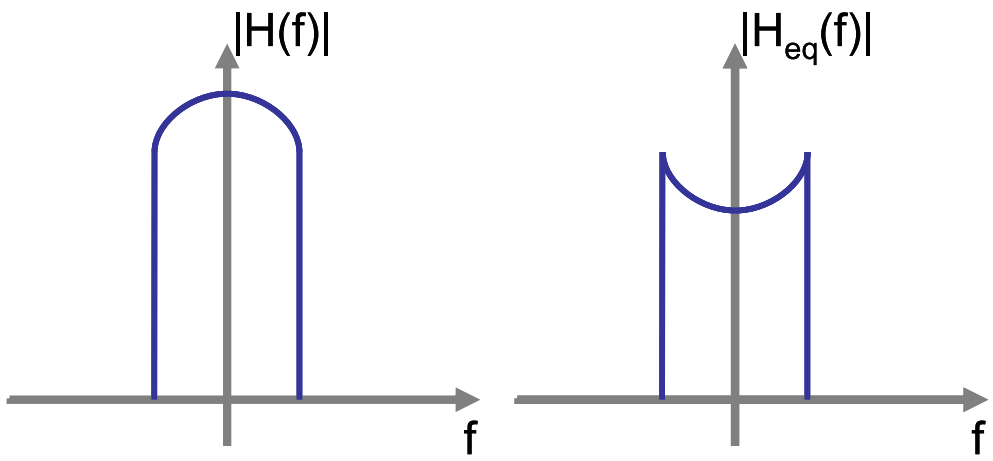
- The analog signal can be recovered from the flat-top PAM signal by the use of a low-pass filter.
- In this case, the restored spectrum $W_r(f)$ is not identical to the spectrum of the original signal $W(f)$ because the high frequency components are attenuated more than lower frequency components.



Instantaneous Sampling (7)

This high-frequency loss can be reduced by:

- Decreasing τ: In this case $H(f) = \tau \mathrm{sinc}(\tau f)$ becomes flatter in the signal band $-B \leq f \leq B$.
  - The disadvantage of this approach is an increase in the bandwidth.
- Equalization: In this case, the loss at high frequencies can be compensated by using a low-pass adapted filter that introduces more gain at higher frequencies.
  - In other words, the transfer function of this equalization filter is $H_{eq}(f) = \frac{1}{H(f)}$ for $-B \leq f \leq B$.



Disadvantages of PAM

The disadvantages of both naturally-sampled and instantaneously-sampled PAM are:

- The bandwidth required is much larger than that of the original analog signal.
- The noise performance is poor.
  - This renders PAM not suitable for long-distance transmissions.

In general, PAM is used since it provides a means for converting an analog signal to a PCM signal as will be seen in the next lecture.